

Computer Organization and Architecture		
Lab #:	4	
Lab Title:	Decimal to Binary conversion in MIPS	
Submitted by:		
Names		Registration #
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Rubrics name & number					Marks	
					In-Lab	Post-Lab
Engineering Knowledge	R2: Use of Engineering Knowledge and follow Experiment Procedures: Ability to follow experimental procedures, control variables, and record procedural steps on lab report.					
Problem Analysis	R6: Experimental Data Analysis: Ability to interpret findings, compare them to values in the literature, identify weaknesses and limitations.					
Design	R8: Best Coding Standards: Ability to follow the coding standards and programming practices.					
Modern Tools Usage	R9: Understand Tools: Ability to describe and explain the principles behind and applicability of engineering tools.					
Individual and Teamwork	R12: Individual Work Contributions: Ability to carry out individual responsibilities.					
	R13: Management of Team Work: Ability to appreciate, understand and work with multidisciplinary team members.					
Rubrics #	R2	R6	R8	R9	R12	R13
In Lab						
Post- Lab						

LAB: 04

Task No: 01

Decimal to Binary conversion in MIPS.

Code:

.data

prompt: .asciiz "Enter a decimal number: "

result: .asciiz "The binary representation is: "

newline: .asciiz "\n"

space: .asciiz " "

.text

main:

Prompt the user for a decimal number

li \$v0, 4 # syscall to print string

la \$a0, prompt # load address of prompt

syscall

Read the decimal number

li \$v0, 5 # syscall to read integer

syscall

move \$t0, \$v0 # store the decimal number in \$t0

Print the result message

li \$v0, 4 # syscall to print string

la \$a0, result # load address of result

syscall

Initialize variables

li \$t1, 32 # bit counter

Convert_loop:

```

beqz $t1, end_print  # if the bit counter is 0, end print
srl $t2, $t0, 31      # shift right to get the leftmost bit
andi $t2, $t2, 1      # mask all bits except the last one
addi $t2, $t2, '0'    # convert bit to ASCII '0' or '1'
li $v0, 11            # syscall to print character
move $a0, $t2         # move bit to $a0
syscall
sll $t0, $t0, 1       # shift left the number by 1 bit to process next bit
subi $t1, $t1, 1      # decrement the bit counter
j Convert_loop        # repeat the loop

```

end_print:

```

# Print a newline
li $v0, 4             # syscall to print string
la $a0, newline
syscall

```

Exit the program

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li $v0, 10           # syscall to exit
syscall

```

Output:

Mars Messages	Run I/O
<pre> Enter a decimal number: 9 The binary representation is: 00000000000000000000000000001001 -- program is finished running -- Enter a decimal number: 4 The binary representation is: 0000000000000000000000000000100 -- program is finished running -- </pre>	